FINAL PROJECT REPORT

LPG Gas Leakage Detection and Automatic Cut Off System

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Bachelor of Technology in Electronics and Communication Engineering

Submitted by

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Abstract

1. Introduction

LPG is a mixture mainly of propane and butane, stored under pressure as a liquid and supplied in cylinders. It is colourless and, in its pure form, odourless, so suppliers add a strong-smelling compound called ethyl mercaptan to make leaks noticeable. Even so, people may not detect a leak in time, particularly when they are asleep, away from home, or suffer from a reduced sense of smell. LPG is heavier than air and tends to settle near the floor, where it can accumulate unnoticed until a spark or flame ignites it.

Accidents caused by gas leaks continue to be reported regularly, and most of them could have been avoided with early warning and automatic intervention. Advances in low-cost semiconductor gas sensors and microcontrollers now make it practical to build an affordable electronic safety device that watches for leaks around the clock and acts without waiting for a human response. This project is developed with that goal in mind.

2. Problem Statement and Objectives

Conventional reliance on smell and manual vigilance is not dependable for LPG safety. The problem addressed here is the lack of an inexpensive system that can detect an LPG leak in real time, alert the occupants, and automatically stop the gas flow before a hazardous concentration builds up.

The main objectives of the project are to detect LPG concentration in air continuously using an MQ-series sensor; to compare the measured value with a safe threshold using a microcontroller programmed in Embedded C; to give an immediate audible and visual warning through a buzzer and LED; to cut off the gas supply automatically when a leak is confirmed; and to provide optional remote alerts through GSM or Bluetooth so that the user can be informed even when away from the premises.

3. System Overview

The system follows a simple sense, decide and act architecture. The gas sensor acts as the input stage, the microcontroller is the decision unit, and the output stage consists of the local alarm, the cut off actuator and the optional communication module. Figure 1 shows the overall block diagram.

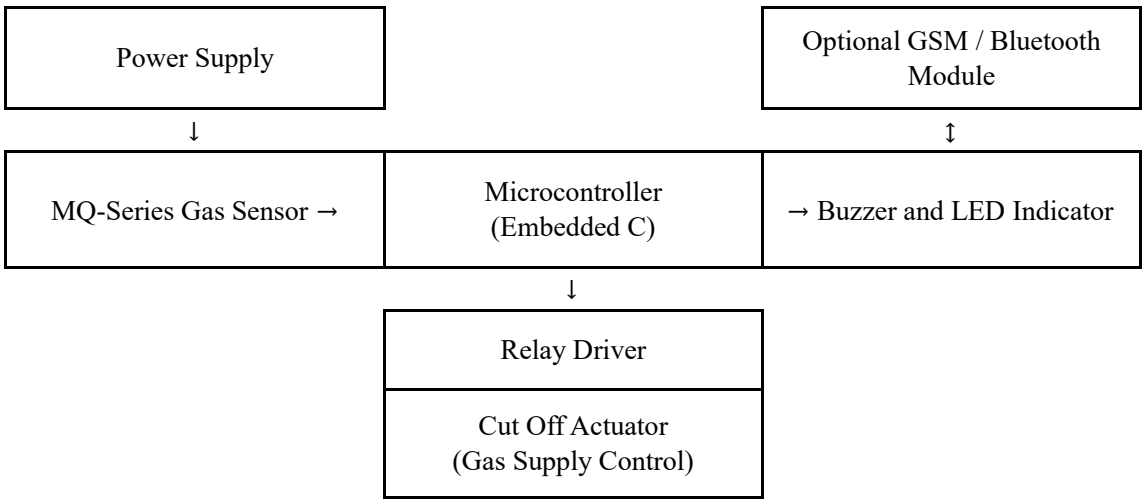


Figure 1: Block diagram of the LPG leakage detection and cut off system (signal flow: sensor to microcontroller to outputs)

4. Hardware Description

The hardware is built from a small number of readily available components, which keeps the cost low and the circuit easy to assemble and maintain. The role of each component is summarised in Table 1.

Component	Function
MQ-series gas sensor (e.g. MQ-6 / MQ-2)	Senses LPG and related combustible gases and outputs a voltage that rises with gas concentration.
Microcontroller	Reads the sensor output, compares it with the threshold and controls all outputs.
Buzzer and LED	Provide an audible and visual alarm at the site of the leak.
Relay and cut off actuator	Switches the mechanism that closes the gas supply when a leak is detected.
GSM / Bluetooth module (optional)	Sends an SMS or wireless notification to the user.
Power supply	Provides regulated voltage to the sensor, controller and output devices.

Table 1: Major components and their functions

4.1 MQ-Series Gas Sensor

MQ-series sensors use a tin dioxide (SnO_2) sensing layer whose electrical conductivity increases when combustible gases are present. A small internal heater keeps the layer at its operating temperature. The sensor provides an analog output proportional to gas concentration, and many modules also include a comparator with a digital output whose trigger level can be adjusted with a potentiometer. The sensor needs a warm-up period after power-on before readings become stable, and this is handled in the firmware.

4.2 Microcontroller

The microcontroller samples the analog sensor output through its analog-to-digital converter and executes the decision logic. Its digital pins drive the buzzer, the LED and the relay, while a serial interface connects the optional GSM or Bluetooth module.

4.3 Alert and Cut Off Stage

The buzzer and LED give an immediate local warning. For the cut off function, the microcontroller drives a relay through a transistor driver stage; the relay in turn operates the shut-off mechanism on the gas line. Using a relay isolates the low-voltage control circuit from the actuator and protects the microcontroller from switching transients.

5. Software Design

The firmware is written in Embedded C. After initialising the input and output pins, the ADC and the serial interface, the program waits for the sensor warm-up period. It then enters a continuous loop in which it reads the sensor value, compares it with the threshold, and either keeps the outputs inactive or starts the alert sequence. A short confirmation check, in which the value must stay above the threshold for a few consecutive samples, helps prevent false alarms caused by momentary disturbances such as cooking vapours or aerosol sprays. The sequence of operation is as follows.

Step 1: Initialise ports, ADC and communication interface. **Step 2:** Allow the sensor to warm up. **Step 3:** Read the analog value from the sensor. **Step 4:** Compare the reading with the threshold. **Step 5:** If the reading is below the threshold, keep the buzzer, LED and relay inactive and return to Step 3. **Step 6:** If the reading stays above the threshold, switch on the buzzer and LED, operate the relay to cut off the gas supply, and send an alert message if the GSM or Bluetooth module is fitted. **Step 7:** Continue monitoring and restore normal operation only after the gas level falls below the threshold and the system is reset.

6. Working Principle

Under normal conditions the sensor reports a low value and the system stays in standby with the alarm and actuator off. When LPG leaks and spreads into the area around the sensor, the sensing element’s conductivity rises and the output voltage increases. As soon as the microcontroller finds that this value has crossed the threshold, it treats the situation as a leak. The buzzer sounds and the LED lights up to warn people nearby, and the relay operates the cut off mechanism so that no further gas is released. If the optional communication module is connected, a message is also sent to the registered user. This combination of local alarm, automatic isolation and remote notification shortens the time between the start of a leak and effective action.

7. Applications

The system can be used in domestic kitchens, restaurants and canteens, hostels, LPG storage and distribution points, small industrial units, laboratories and any other place where LPG is stored or used. With suitable sensor selection it can also be adapted to detect other combustible gases.

8. Advantages and Limitations

The main advantages of the system are its low cost, small size, simple circuit, fast response and the fact that it does not depend on human attention to act. The optional GSM or Bluetooth link adds remote awareness without changing the core design. The limitations are those common to semiconductor sensors: the sensor responds to other gases and vapours as well as LPG, its readings drift with temperature and humidity, it needs periodic calibration, and

its heater draws a noticeable current. The GSM option depends on network coverage, and the Bluetooth option works only within a short range.

9. Future Scope

The design can be extended in several directions. Internet of Things connectivity would allow data logging and alerts through a mobile application. An exhaust fan could be switched on automatically to disperse the gas, and a second sensor or a temperature and humidity sensor could be added to improve reliability and compensate for drift. A battery backup would keep the system working during power failures, and a motorised valve could provide a more robust shut-off than a relay-driven mechanism alone.

10. Conclusion

This project demonstrates a practical and affordable LPG leakage detection and automatic cut off system using an MQ-series sensor and a microcontroller programmed in Embedded C. By combining continuous monitoring, immediate audible and visual alerts, automatic gas shut-off and optional remote notification, the system detects threshold breaches early and helps reduce fire hazards in domestic and industrial settings. The design is simple enough to build and maintain at low cost, and it offers a solid base for further enhancements such as IoT integration and improved sensor compensation.

References

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